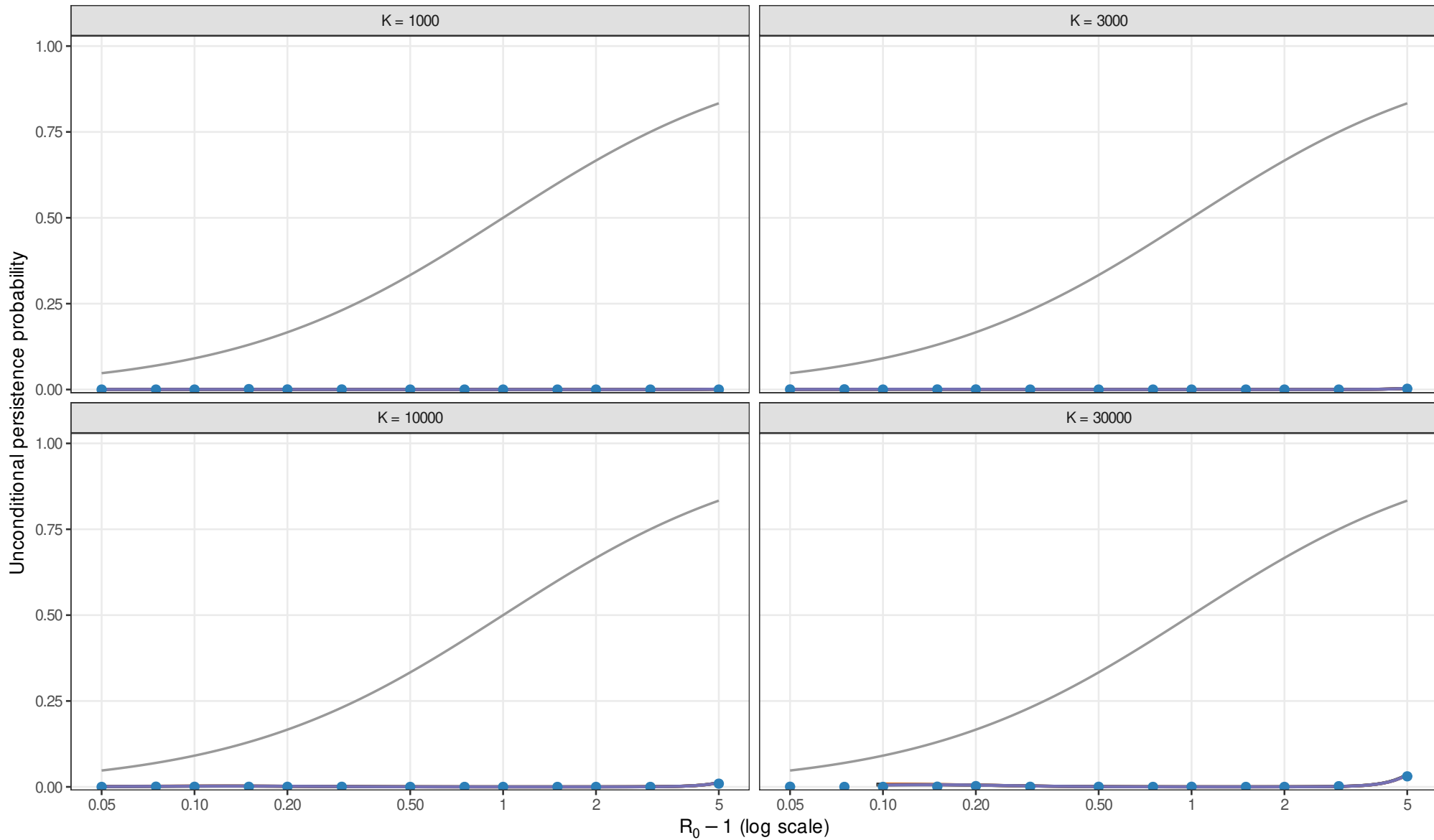


Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.01$, $\theta = 0$; $l_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

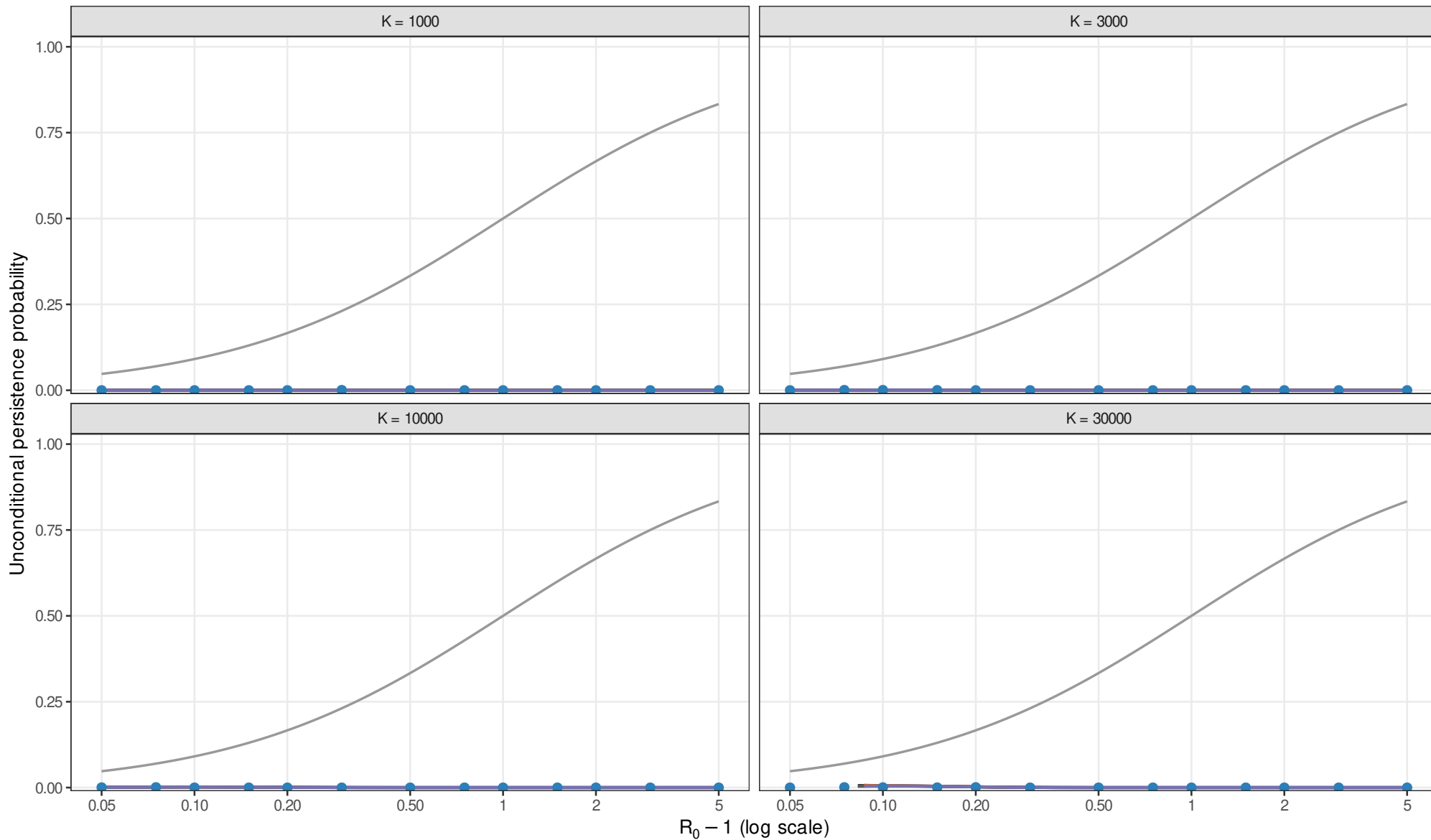
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.01$, $\theta = 0.5$; $I_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

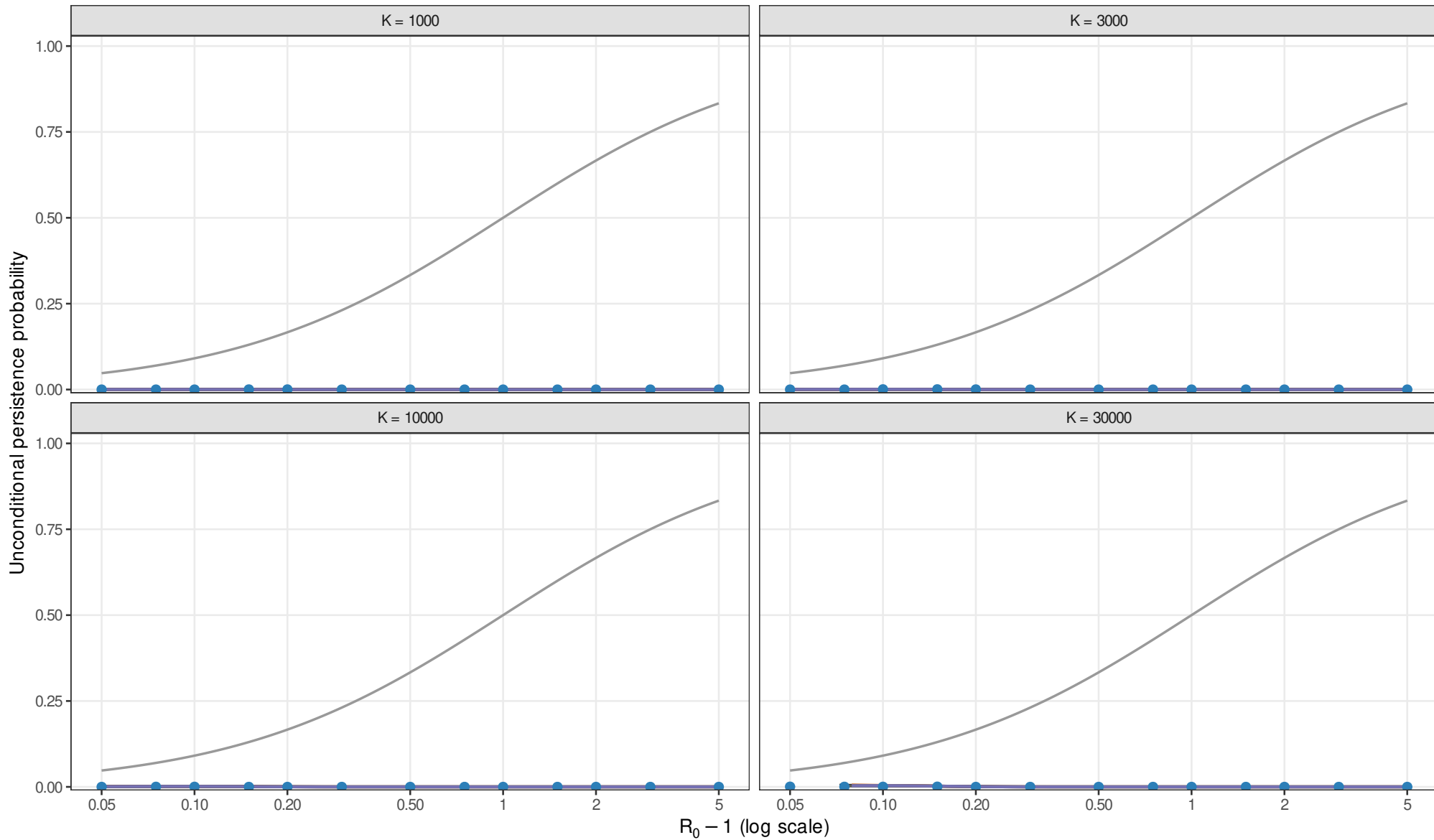
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.01$, $\theta = 1$; $l_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

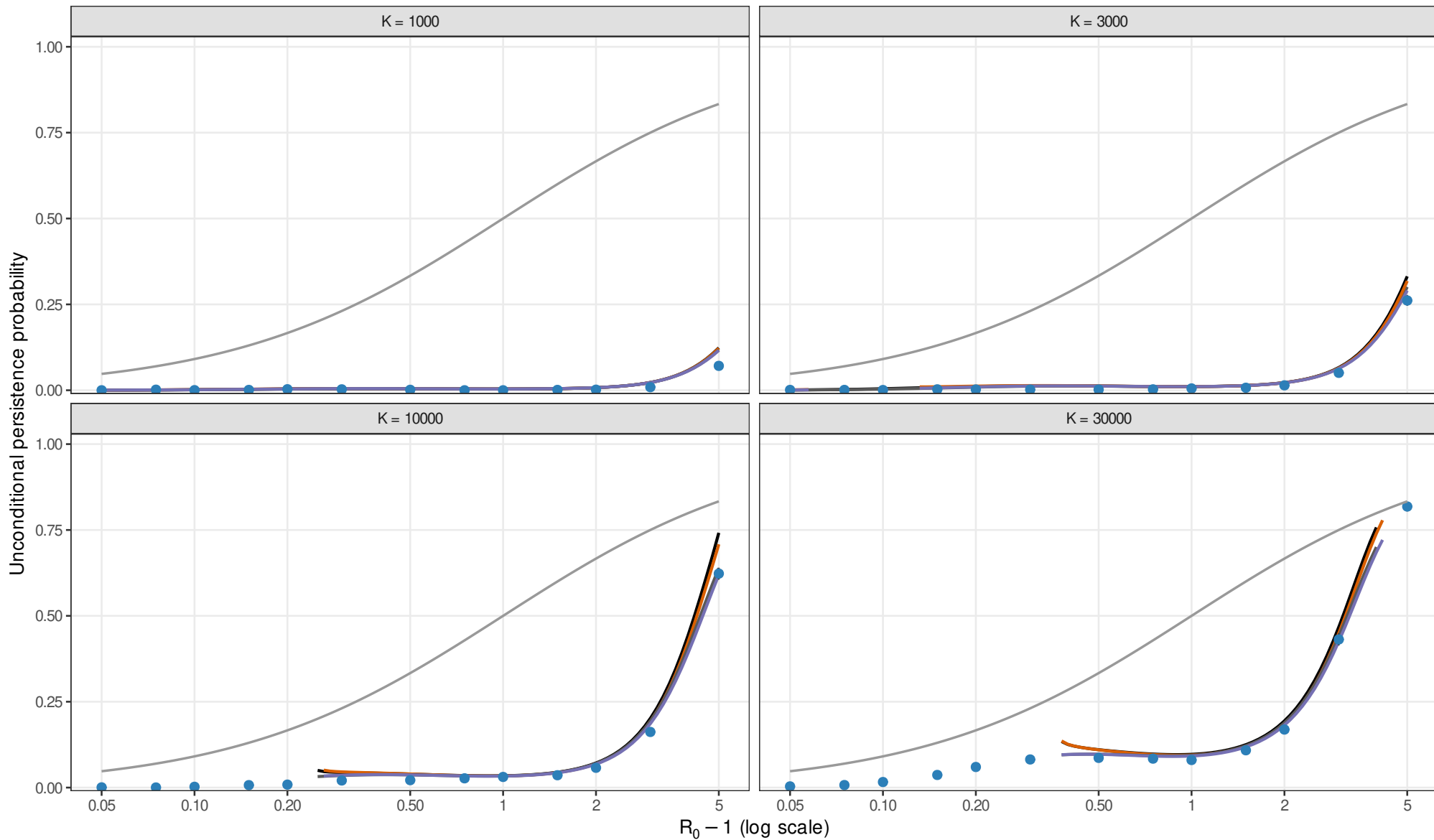
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 0$; $l_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

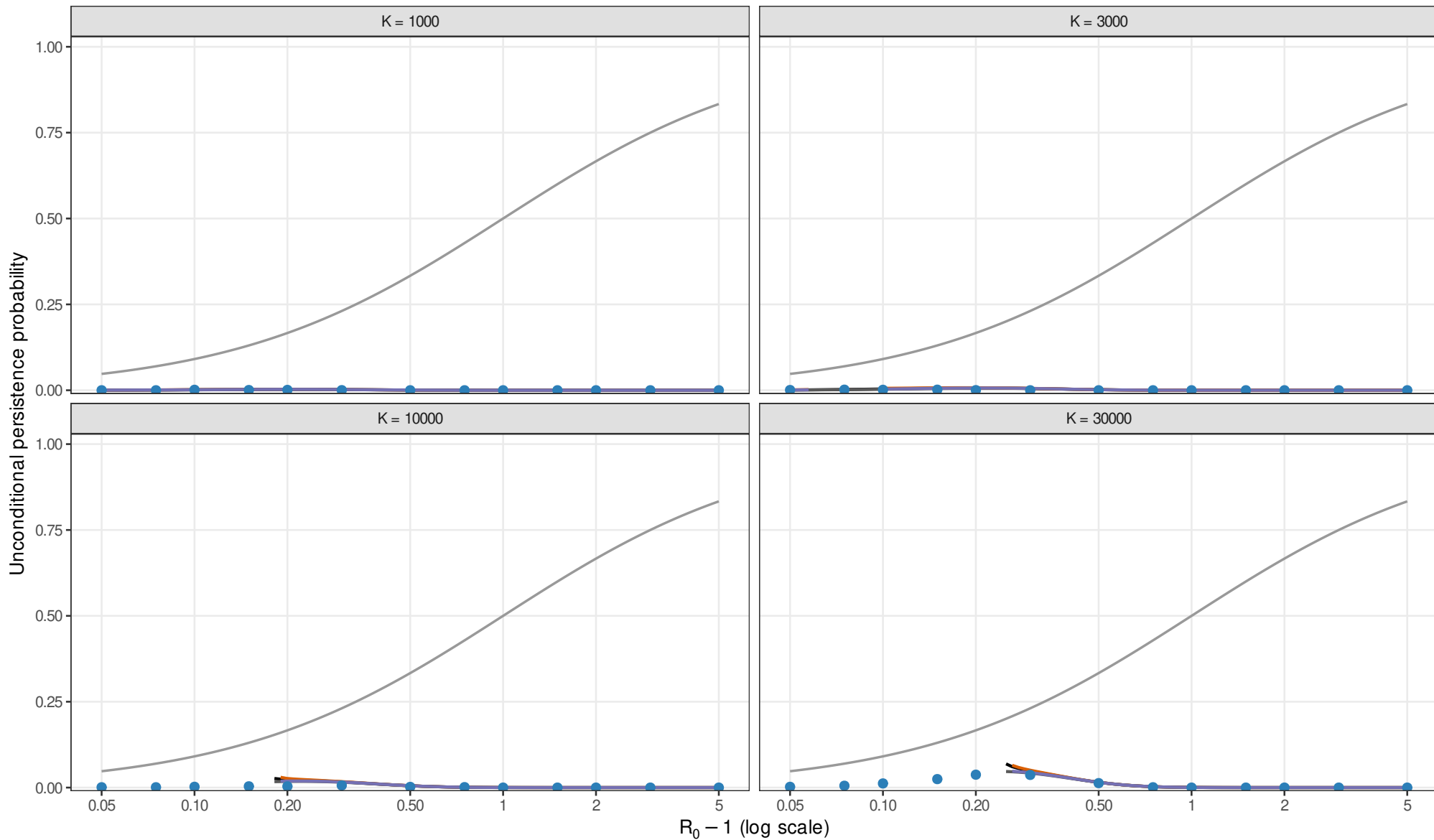
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 0.5$; $I_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

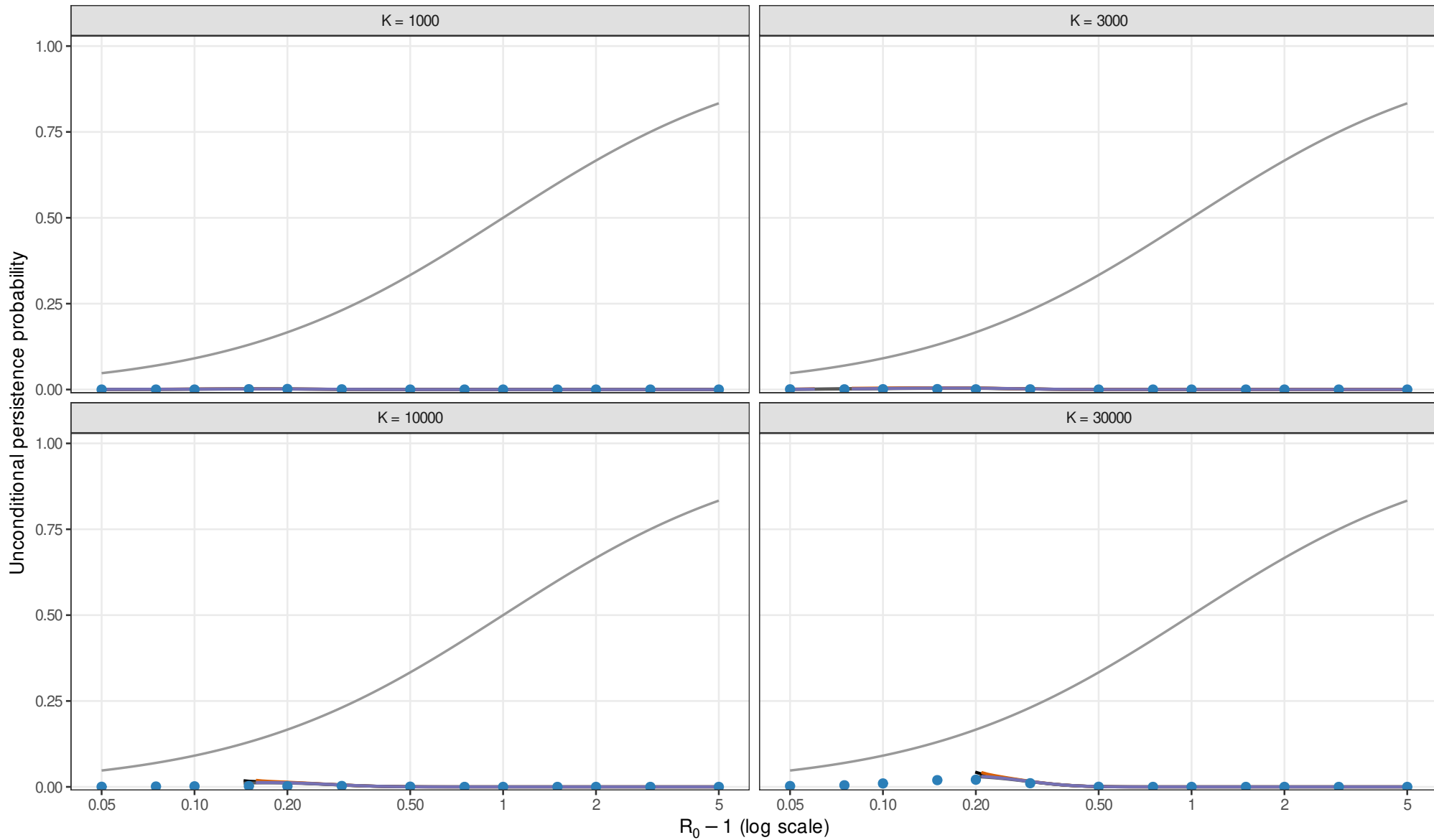
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 1$; $l_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

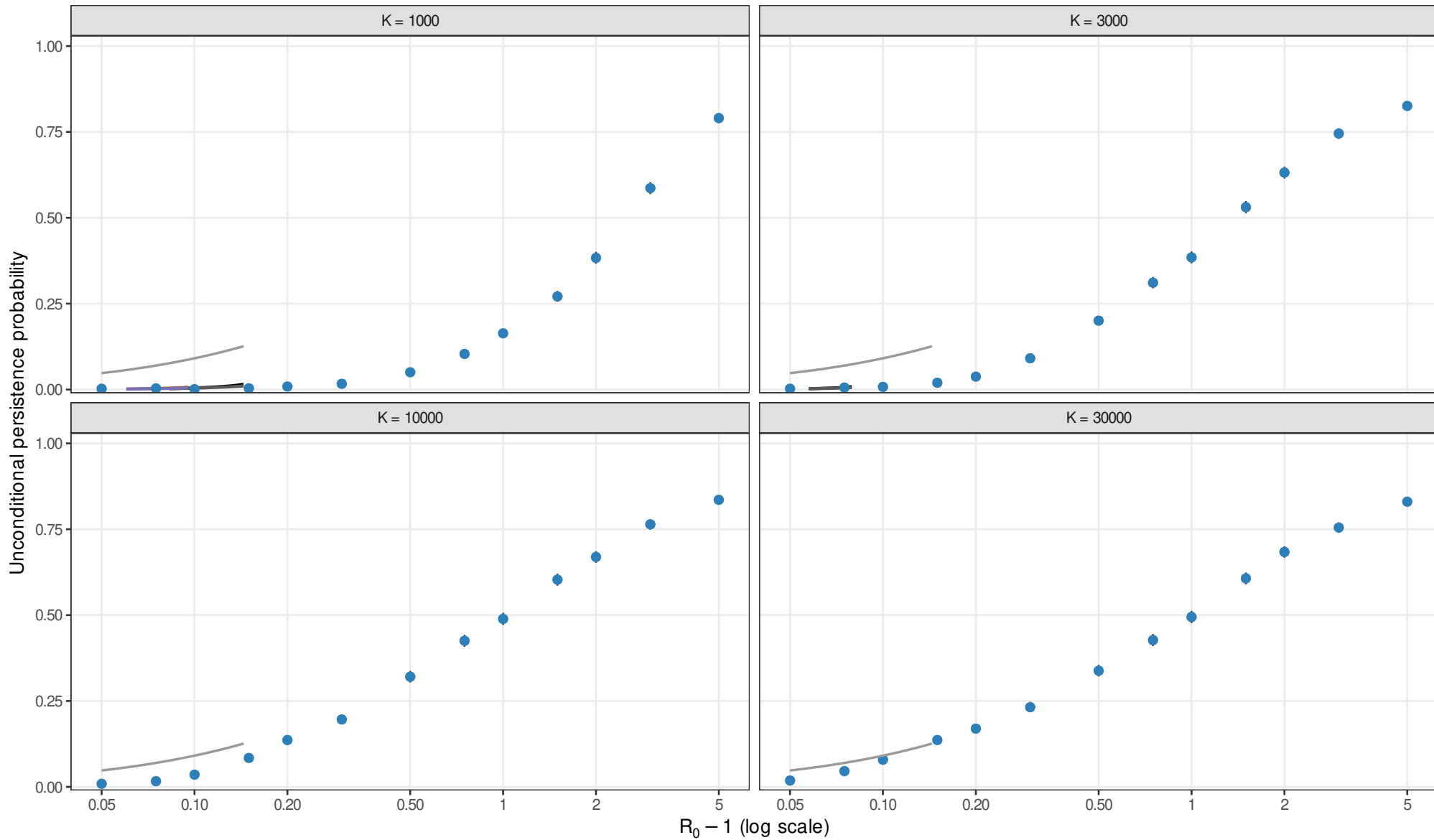
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$.
Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.05, theta = 0; l0 = 1; matching height = sqrt(y*/K)



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

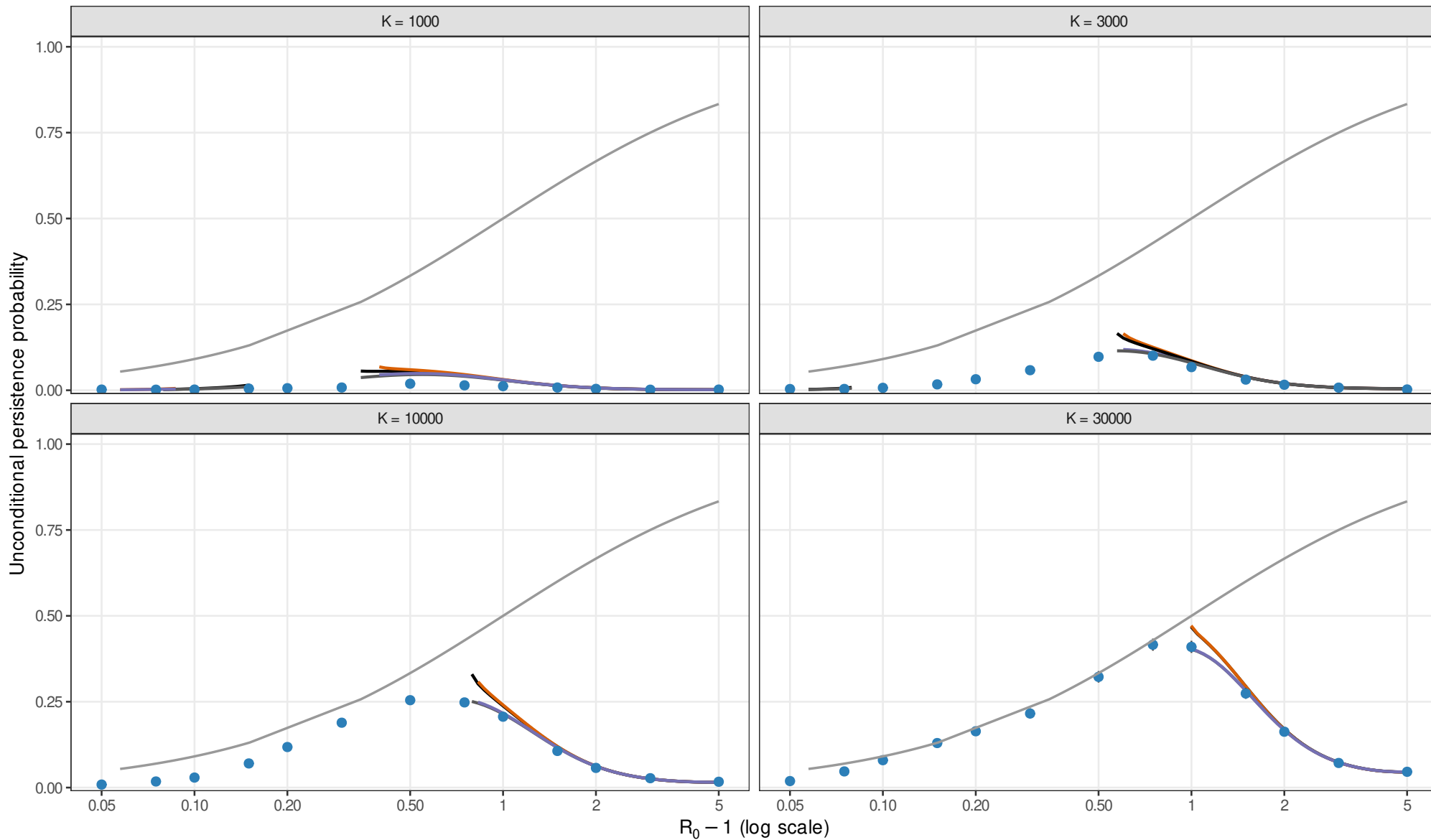
Matching height

- $y_{BL} = \sqrt{y_{star}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.05$, $\theta = 0.5$; $I_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

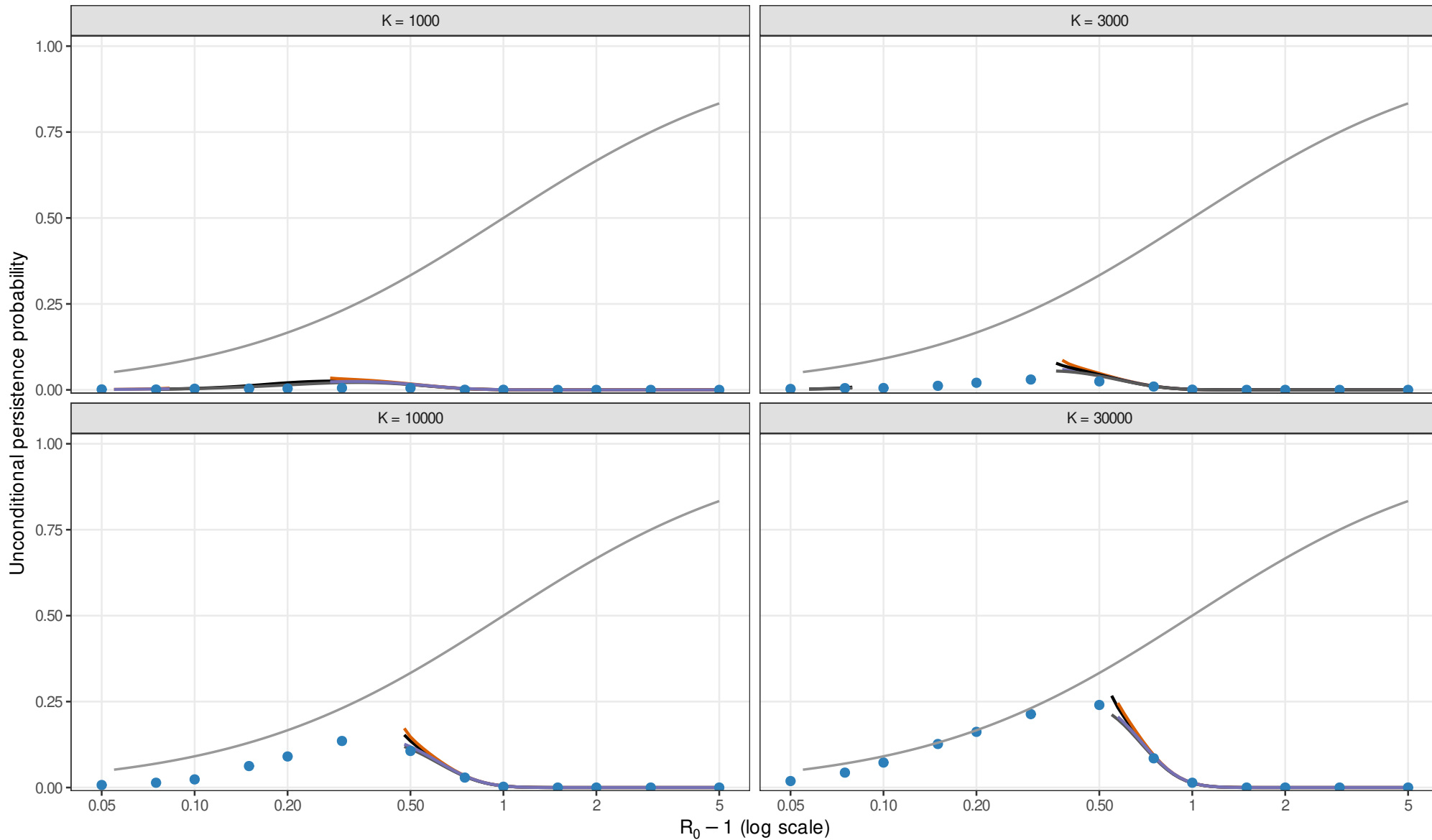
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.05$, $\theta = 1$; $l_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

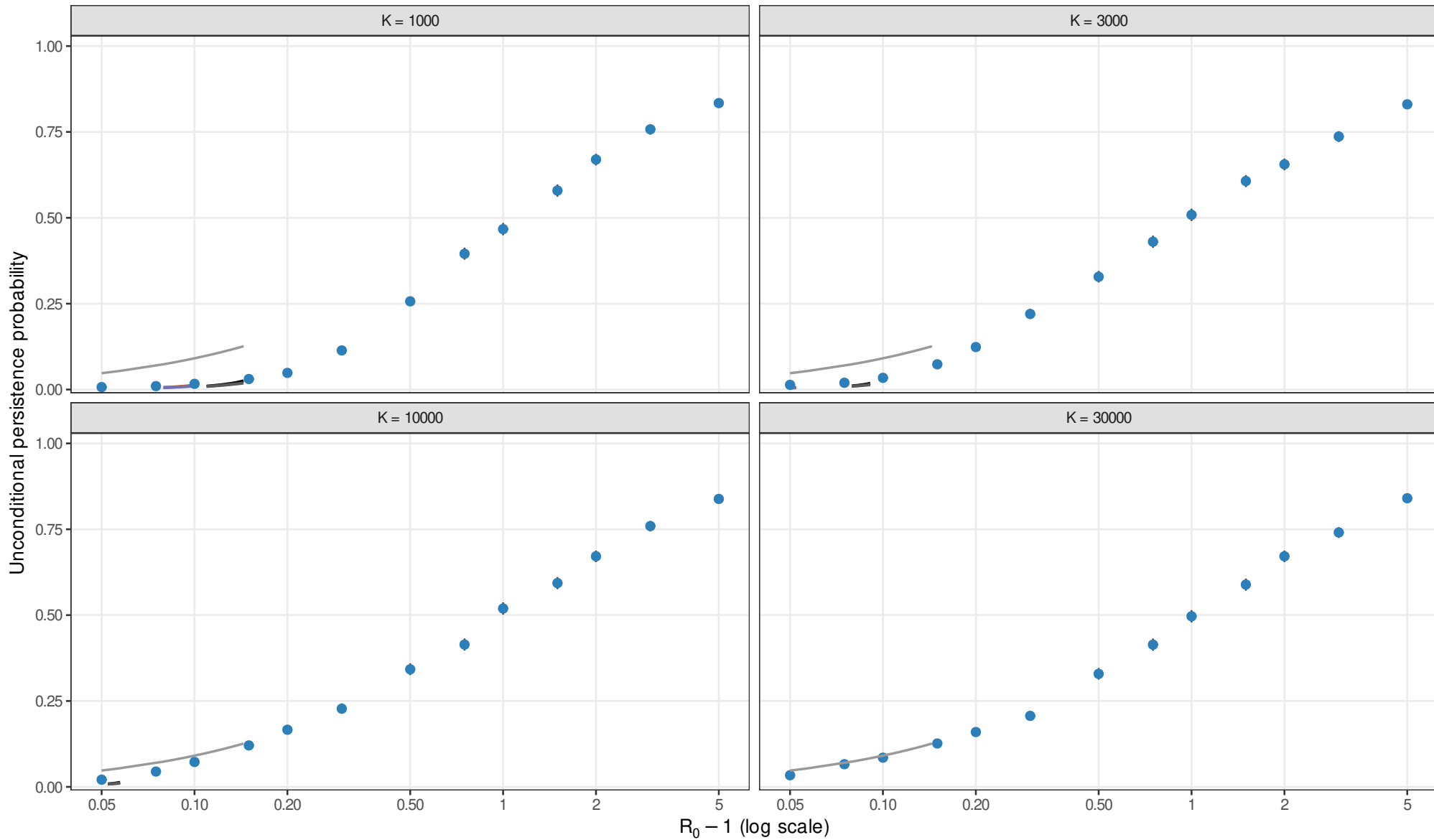
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.10$, $\theta = 0$; $l_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

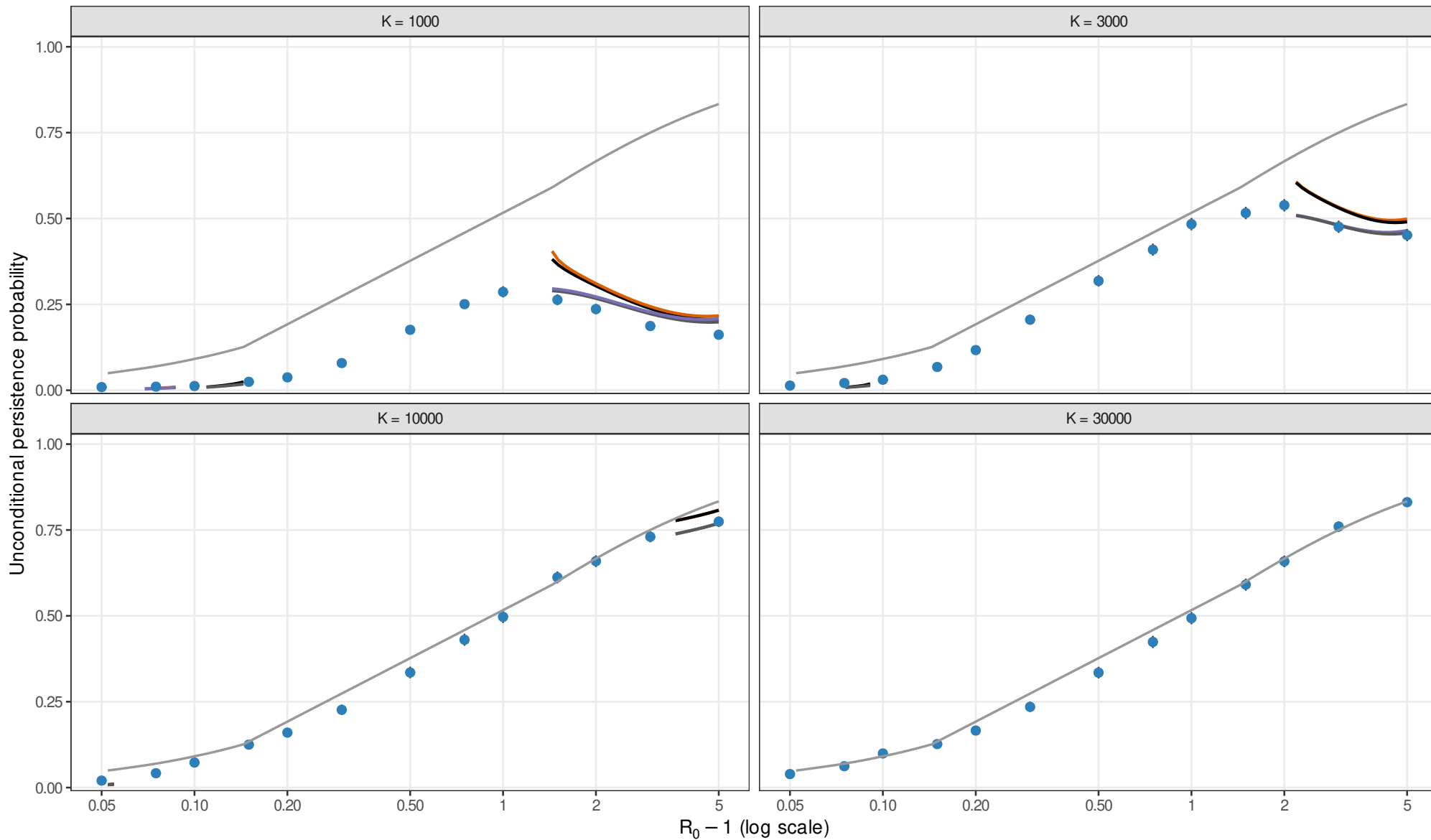
Matching height

- $y_{BL} = \sqrt{y_{\text{star}}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.10$, $\theta = 0.5$; $I_0 = 1$; matching height = $\sqrt{y^*/K}$



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

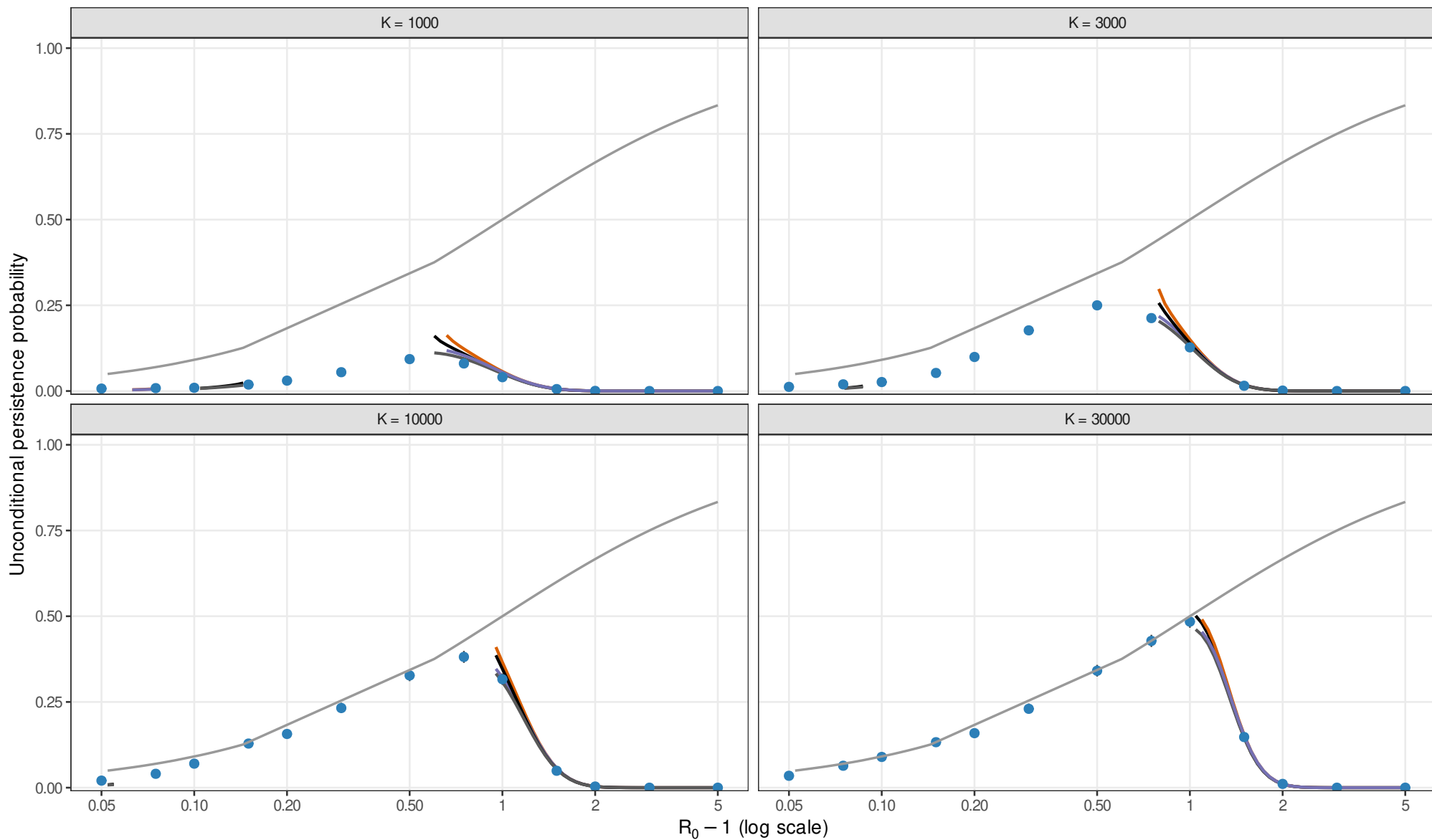
Matching height

- $y_{BL} = \sqrt{y_{star}/K}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

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Analytical method

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Matching height

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Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $\sqrt{y^*/K}$. Each curve stops where its matching equation has no admissible root.